

MA341

APPLIED ODE's

Homework 1.5

1. Determine the order of the following DEs and whether it is linear or not.

(a) $y^{(3)} + 2x = y$

(b) $y'' + 2y = \frac{1}{y} + \pi$

(c) $(x + y')^2 = (x - y')^2 + 1$

2. Find a general solution to the following ODEs:

(a) $y' = \frac{x^2}{x^2+1}$

(b) $(x^2 + 1)y' + xy = x$

(c) $y' = (\sin(x) + 1)y$

(d) $\frac{dr}{d\theta} + r \tan \theta = \frac{1}{\cos \theta}$

3. Find solution to the following initial value problems

(a) $\begin{cases} \frac{dy}{dx} - \frac{y}{x} = xe^x \\ y(1) = e - 1 \end{cases}$

(b) $\begin{cases} t^3 \frac{dx}{dt} + 3t^2 x = t \\ x(2) = 0 \end{cases}$

4. Solve the following Bernoulli equations

(a) $xy' + y = y^2 \ln x$

(b) $y' - y = x\sqrt[3]{y}$

5. Solve Riccati equation $y' + 2ye^x - y^2 = e^{2x} + e^x$, given a private solution $y = e^x$.